

Jack Hesse-Withbroe

101 Roosevelt Ave • Eau Claire, WI 54703 • JHWithbroe@outlook.com • [Digital Portfolio](#)

Education

University of Wisconsin-Eau Claire

Graduation: May 2026

- **Degree:** B.S Geospatial Analysis & Technology | **Certificates:** Information Systems, Earth Resources

Technical Skills

Software Programs

- **ArcGIS Pro:** Advanced geoprocessing Tools, Spatial Analyst, Network Analyst, Business Analyst, ArcPy, Interpolation, ModelBuilder, Arcade Expressions, Advanced layout design.
- **ArcGIS Online, ArcGIS Enterprise:** Webmapping, Experience Builder, StoryMaps, Dashboards
- **Data Management:** Enterprise geodatabases, schema consistency, QA/QC, metadata, Experience working with feature classes, shapefiles, hosted feature services, and spatial joins
- **ESRI FieldMaps, QGIS, Survey 123, ArcMap, ArcScene, Erdas Imagine**
- **Adobe Illustrator:** Professional cartographic refinement, report-ready map production
- **Microsoft Word, PowerPoint, Teams, Excel:** Dataset Creation, Dataset Manipulation

Languages

- **Python:** Geopandas, Pandas, Numpy, Matplotlib, Folium, Rasterio, GDAL, Shapely, ArcPy
- **R:** TidyVerse, TidyCensus, Leaflet, DPLYR
- **HTML, JavaScript, SQL**

Other

- **Field work:** GPS equipment & data collection

Relevant Work Experience

G.I.S Intern – Washington County Government

Oak Park Heights, MN June 2025-December 2025

- Maintained and updated enterprise GIS datasets supporting county planning, infrastructure, and land info systems.
- Processed, edited, and validated spatial data including transportation and trails layers, ensuring attribute accuracy and spatial integrity.
- Performed structured QA/QC to resolve projection discrepancies, geometry errors, and schema inconsistencies.
- Interpreted plats, aerial imagery, and source documentation to support accurate feature representation.
- Assisted with addressing and road centerline dataset maintenance and supported internal GIS troubleshooting requests.
- Documented workflows and maintained records to ensure consistent data standards and long-term system reliability.

Relevant Coursework

Geospatial

- G.I.S I, G.I.S II, G.I.S III, Quantitative Methods in Geography, Geospatial Field Methods, Applied Cartography and Geovisualization, Business G.I.S, Advanced Remote Sensing & Data Analytics, Remote Sensing of the Environment, Introduction to Geospatial Analysis, Cartography

Scripting/Programming

- Computing with Python, Data Analysis with R, Computing with JavaScript, Webpage Design with HTML

Environmental

- Biogeography, Climatology, Conservation of the Environment, Physical Geography

Leadership Experience

Captain - University of Wisconsin-Eau Claire Cross Country, Track & Field teams

Fall 2023-Spring 2026

- Year round committed approximately 30 hours per week to training, traveling across country to meets, managing team administration, and leading team meetings all while maintaining full course load and employment

